

Oracle Datafile Media Recovery VS Block Media Recovery



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Performing Block Media Recovery

This topic explains how to restore and recover individual data blocks within a data file and we following these topics:

- Overview of Block Media Recovery
- Prerequisites for Block Media Recovery
- Recovering Individual Blocks
- Recovering All Blocks in
V\$DATABASE_BLOCK_CORRUPTION

Overview of Block Media Recovery

Block media recovery recovers provides lower mean time to recover (MTTR) by recovering corrupt data blocks.

This section contains the following topics:

- Purpose of Block Media Recovery
- Basic Concepts of Block Media Recovery

Purpose of Block Media Recovery

Use block media recovery to recover one or more corrupt data blocks within a data file.

Block media recovery provides the following advantages over data file media recovery:

- Lowers the mean time to recover (MTTR) because only blocks needing recovery are restored and recovered
- Enables affected data files to remain online

during recovery

Without block media recovery, if even a single block is corrupt, then you must take the data file offline and restore a backup of the data file. You must apply all redo generated for the data file after the backup was created. The entire file is unavailable until media recovery completes. With block media recovery, only the blocks actually being recovered are unavailable during the recovery.

Block media recovery is most useful for physical corruption problems that involve a small, well-known number of blocks. Block-level data loss usually results from intermittent, random I/O errors that do not cause widespread data loss, and memory corruptions that are written to disk. Block media recovery is not intended for cases where the extent of data loss or corruption is unknown and the entire data file requires recovery. In such cases, data file media recovery is the best solution.



Basic Concepts of Block Media Recovery

Usually, the database marks a block as media corrupt and then writes it to disk when the corruption is first encountered. No subsequent read of the block is successful until the block is recovered. You can perform block recovery only on blocks that are marked corrupt or that fail a corruption check.

Typically, block corruption is reported in the following locations:

- Results of the LIST FAILURE, VALIDATE, or BACKUP ... VALIDATE command
- The V\$DATABASE_BLOCK_CORRUPTION view
- Error messages in standard output
- The alert log
- User trace files

- Results of the SQL commands ANALYZE TABLE and ANALYZE INDEX
- Results of the DBVERIFY utility
- Third-party media management output

For example, you may discover the following messages in a user trace file:

ORA-01578: ORACLE data block corrupted (file # 7, block # 3)

ORA-01110: data file 7: '/oracle/oradata/trgt/tools01.dbf'

ORA-01578: ORACLE data block corrupted (file # 2, block # 235)

ORA-01110: data file 2: '/oracle/oradata/trgt/undotbs01.dbf'

Note:

Starting in Oracle Database 19c, the Data Recovery Advisor (DRA) feature is **deprecated**.

The deprecation of DRA includes deprecation of the following Oracle Recovery Manager (RMAN) commands:

LIST FAILURE, ADVISE FAILURE, REPAIR FAILURE, and CHANGE FAILURE.

Database administrators will no longer have access to these commands. There is no replacement feature for DRA.

About Block Recovery and Standby Databases

Block recovery behavior depends on whether the data block corruption was discovered on the primary database or the physical standby database.

If the database on which the corruption occurs is associated with a real-time query physical standby database, then the database automatically attempts to perform block media recovery.

The primary database searches for good copies of blocks on the standby database and, if they are found, repairs the blocks with no impact to the query that encountered the corrupt block.

The Oracle Database physical block corruption message (ORA-1578) is displayed only if the database cannot repair the corruption.

Whenever block corruption has been automatically detected, you can perform block media recovery manually with the RECOVER ... BLOCK command.



By default, RMAN first searches for good blocks in the real-time query physical standby database, then flashback logs and then blocks in full or level 0 incremental backups.

Note:

For block media recovery to work automatically, the physical standby database must be in real-time query mode.

An Oracle Active Data Guard license is required.

If a corrupt data block is discovered on a real-time query physical standby database, the server

attempts to repair the corruption by obtaining a copy of the block from the primary database. The repair is performed in the background, enabling subsequent queries to succeed if the repair is successful.

Automatic block repair is attempted if the following database initialization parameters are configured on the standby database as described:

- The LOG_ARCHIVE_CONFIG parameter is configured with a DG_CONFIG list and a LOG_ARCHIVE_DEST_n parameter is configured for the primary database with the DB_UNIQUE_NAME attribute

or

- The FAL_SERVER parameter is configured and its value contains an Oracle Net service name for the primary database



Note:

If a corrupt block is detected during validation, such as by the RMAN VALIDATE command, then recovery is not initiated automatically.

About Identifying Corrupt Blocks

The V\$DATABASE_BLOCK_CORRUPTION view displays blocks marked corrupt by database components such as RMAN, ANALYZE, and SQL queries.

The following types of corruption result in the addition of rows to this view:

- **Physical corruption (sometimes called media corruption)**

The database does not recognize the block: the checksum is invalid, the block contains all zeros, or the block header is corrupt.

Physical corruption checking is enabled by default.

You can turn off checksum checking by specifying the **NOCHECKSUM** option of the **BACKUP** command, but other physical consistency checks, such as checks of the block headers and footers, **cannot be disabled**.

- **Logical corruption**

The block has a valid checksum, the header and footer match, and so on, but the contents are logically inconsistent. Block media recovery may not be able to repair all logical block corruptions.

In these cases, alternate recovery methods, such as tablespace point-in-time recovery, or dropping and re-creating the affected objects, may repair the corruption.

Logical corruption checking is disabled by default.

You can turn it on by specifying the **CHECK LOGICAL** option of

the **BACKUP, RESTORE, RECOVER,**
and **VALIDATE** commands.

The database can detect some corruptions by validating relationships between blocks and segments, but cannot detect them by a check of an individual block.

The `V$DATABASE_BLOCK_CORRUPTION` view does not record at this level of granularity.



About Missing Redo During Block Recovery

Block media recovery only requires an unbroken set of redo changes for the blocks being recovered. This is unlike data file recovery that requires an unbroken series of redo changes from the beginning of recovery to the end.

Like data file media recovery, block media recovery cannot generally survive a missing or inaccessible

archived log, although it attempts restore failover when looking for usable copies of archived redo log files. Also, block media recovery cannot survive physical redo corruptions that result in checksum failure.

However, block media recovery can survive gaps in the redo stream if the missing or corrupt redo records do not affect the blocks being recovered.

Note:

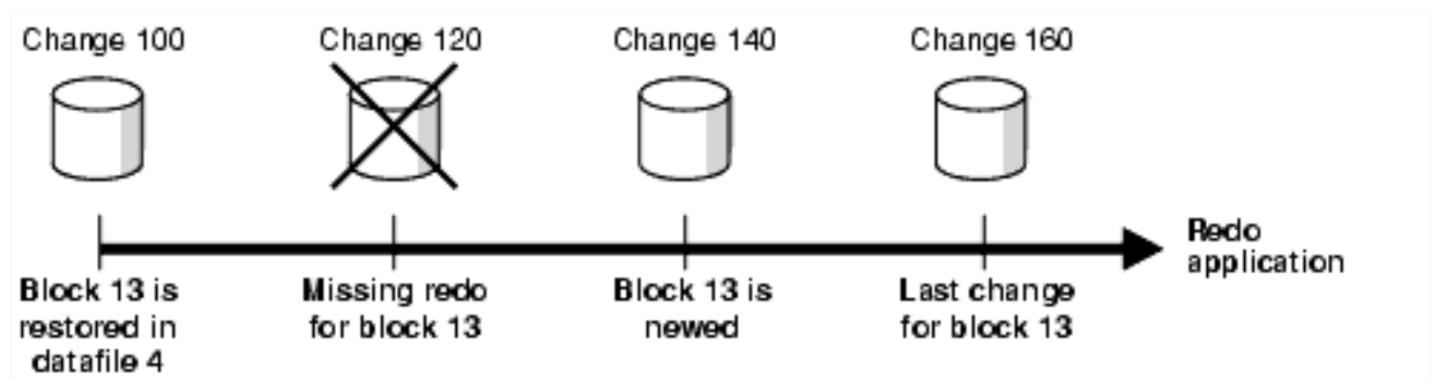
Each block is recovered independently during block media recovery, so recovery may be successful for a subset of blocks.

When RMAN first detects missing or corrupt redo records during block media recovery, it does not immediately signal an error because the block undergoing recovery may create one later in the redo stream.

When a block is re-created, all previous redo for that block becomes irrelevant because the redo applies to an old incarnation of the block.

For example, the database creates a new a block when users drop or truncate a table and then use the block for other data.

Assume that media recovery is performed on block 13 as depicted in below figure:



After block recovery begins, RMAN discovers that change 120 is missing from the redo stream, either because the log block is corrupt or because the log cannot be found.

RMAN continues recovery if block 13 is re-created later in the redo stream.

Assume that in change 140 a user drops the table employees stored in block 13, allocates a new table in this block, and inserts data into the new table.

At this point, the database formats block 13 as a

new block.

Recovery can now proceed with this block even though some redo preceding the recreation operation was missing!



Prerequisites for Block Media Recovery

Certain prerequisites must be met before you perform block media recovery by using the RECOVER ... BLOCK command.

The **prerequisites** include the following:

- The target database must run in **ARCHIVELOG** mode and be open or mounted with a current control file.
- If the target database is a standby database, then it must be in a consistent state, recovery cannot be in session, and the backup must be

older than the corrupted file.

- The backups of the data files containing the corrupt blocks must be full or level 0 backups. They cannot be proxy copies or incremental backups.

If only proxy copy backups exist, then you can restore them to a nondefault location on disk, in which case RMAN considers them data file copies and searches them for blocks during block media recovery.

- **RMAN can use only archived redo logs for the recovery.**

RMAN cannot use level 1 incremental backups!

Block media recovery cannot survive a missing or inaccessible archived redo log, although it can sometimes survive missing redo records.

- **Flashback Database must be enabled on the target database for RMAN to search the flashback logs for good copies of corrupt**

blocks.

If flashback logging is enabled and contains older, uncorrupted versions of the corrupt blocks, then RMAN can use these blocks, possibly speeding up the recovery.

- **The target database must be associated with a real-time query physical standby database for RMAN to search the database for good copies of corrupt blocks.**



Recovering Individual Blocks

Use the RECOVER...BLOCK command to recover individual corrupt blocks in a data file.

This section contains the following topics:

- **Recovering Individual Blocks Using the RECOVER...BLOCK Command**

- Example: Recovering Individual Blocks Using the Data Recovery Advisor

Recovering Individual Blocks Using the RECOVER...BLOCK Command

You identify the blocks that require recovery and then use any available backup to restore and recover these blocks.

To recover specific data blocks using the RECOVER...BLOCK command:

- Obtain the data file numbers and block numbers of the corrupted blocks.

The easiest way to locate trace files and the alert log is to connect SQL*Plus to the target database and execute the following query:

```
SELECT NAME, VALUE FROM V$DIAG_INFO;
```

- Start RMAN and connect to the target database, which must be mounted or open.

- Run the SHOW ALL command to confirm that the appropriate channels are preconfigured.
- Run the RECOVER ... BLOCK command at the RMAN prompt, specifying the file and block numbers for the corrupted blocks.

The following example recovers two blocks.

```
RECOVER  
DATAFILE 8 BLOCK 13  
DATAFILE 2 BLOCK 19;
```

You can also specify various options to control RMAN behavior.

The following example indicates that only backups with the tag **mondayam** are used when searching for blocks.

You could use the FROM BACKUPSET option to restrict the type of backup that RMAN searches, or the EXCLUDE FLASHBACK LOG option to restrict RMAN from searching the flashback logs.

```
RECOVER  
DATAFILE 8 BLOCK 13  
DATAFILE 2 BLOCK 199  
FROM TAG mondayam;
```

Example: Recovering Individual Blocks Using the Data Recovery Advisor

You can use the Data Recovery Advisor to diagnose and repair failures caused by data block corruptions. In this example, corrupt data blocks were discovered when the `VALIDATE DATABASE` command.

🔴 Starting in Oracle Database 19c, the Data Recovery Advisor (DRA) feature is deprecated.

🙵 The deprecation of DRA includes deprecation of the following Oracle Recovery Manager (RMAN) commands: `LIST FAILURE`, `ADVISE FAILURE`, `REPAIR FAILURE`, and `CHANGE FAILURE`.

Database administrators will no longer have access to these commands. There is no replacement feature for DRA.

To generate automated repair options and repair the failure using the Data Recovery Advisor:

- Start RMAN and connect to the target database.

```
% rman TARGET /  
% rman TARGET user@prod NOCATALOG
```

- List the failures recorded by the Data Recovery Advisor using the following command:



LIST FAILURE;

Database Role: PRIMARY

List of Database Failures

=====

Failure ID	Priority	Status	Time Detected	Summary
-----	-----	-----	-----	-----

5720 HIGH OPEN 24-APR-14 Datafile 14:
'/home1/oracle/dbs/tbs_32.f' contains one or more
corrupt blocks

- Generate repair options for the failure listed
Step 2.

The following command generates repair options
and creates a repair script to perform the
automated repair tasks.

ADVISE FAILURE;

Database Role: PRIMARY

List of Database Failures

=====

Failure ID	Priority	Status	Time Detected	Summary
5720	HIGH	OPEN	24-APR-14	Datafile 14: '/home1/oracle/dbs/tbs_32.f' contains one or more corrupt blocks

analyzing automatic repair options; this may take
some time

using channel ORA_DISK_1

analyzing automatic repair options complete

Mandatory Manual Actions

=====

no manual actions available

Optional Manual Actions

=====

no manual actions available

Automated Repair Options

=====

Option Repair Description

1 Perform block media recovery of block 20 in
file 14

Strategy: The repair includes complete media
recovery with no data loss

Repair script: /home1/oracle/log/diag/rdbms/
db12/hm/reco_287949467.hm

- Perform the automated repairs recommended

by Data Recovery Advisor.

RMAN uses the repair script generated by the ADVISE FAILURE command to perform the required repairs.



REPAIR FAILURE;

Strategy: The repair includes complete media recovery with no data loss
Repair script: /home1/oracle/log/diag/rdbms/db12/hm/reco_287949467.hm
contents of repair script: #
block media recovery recover datafile 14 block 20;

Do you really want to execute the above repair
(enter YES or NO)? yes 🌞

executing repair script Starting recover at 24-
APR-14 using channel ORA_DISK_1 channel
ORA_DISK_1: restoring block(s)channel
ORA_DISK_1: specifying block(s) to restore from
backup setrestoring blocks of datafile 00014

channel ORA_DISK_1: reading from backup piece /
backups/DB121/backupset/2024_04_24/
o1_mf_nnndf_TAG20240424T213309_9omsd7vb_.b
kp channel ORA_DISK_1: piece handle=/backups/
DB121/backupset/2024_04_24/
o1_mf_nnndf_TAG20240424T213309_9omsd7vb_.b
kp tag=TAG20240424T213309 channel
ORA_DISK_1: restored block(s) from backup piece 1
channel ORA_DISK_1: block restore complete,
elapsed time: 00:00:01 starting media recovery
media recovery complete, elapsed time: 00:00:03
Finished recover at 24-APR-14 repair failure
complete

When the LIST FAILURE command displays more than one failures, you can perform repair actions only for a particular failure.

Use the option number displayed in the Automated Repair Options section of the ADVISE FAILURE command output to perform specific repair actions.

The following command performs only the repair actions listed under Option 2 of the Automated

Repair Options section.

REPAIR FAILURE USING ADVISE OPTION 2;

Recovering All Blocks in V\$DATABASE_BLOCK_CORRUPTION

RMAN can automatically recover all blocks listed in the V\$DATABASE_BLOCK_CORRUPTION view.

To recover all blocks logged in
V\$DATABASE_BLOCK_CORRUPTION:

- Start SQL*Plus and connect to the target database.
- Query V\$DATABASE_BLOCK_CORRUPTION to determine whether corrupt blocks exist. For example, execute the following statement:

```
SQL> SELECT * FROM  
V$DATABASE_BLOCK_CORRUPTION;
```

- Start RMAN and connect to the target database, as described in "Making Database Connections

with RMAN".

- Recover all blocks marked corrupt in V\$DATABASE_BLOCK_CORRUPTION.

The following command repairs all physically corrupted blocks recorded in the view:

```
RMAN> RECOVER CORRUPTION LIST;
```

After the blocks are recovered, the database removes them from V\$DATABASE_BLOCK_CORRUPTION.



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